CI/CD Pipeline - GitHub Actions

GitHub Repo
Credit Card
Recommendation

1. Push Code

Pipeline Stages

Stage 1:
Lint & Test

Stage 2:
Security Scan
(Trivy)

3. Security OK

Stage 3:
Docker Build

4. Build Docker

Stage 4:
Push to ECR

2. Tests Pass

Stage 5:
Deploy to EKS

Notify

Slack
Notification

7. Deploy

End Users
(Global)

HTTPS Traffic

AWS Cloud - us-east-1 Region

Container Registry

Amazon ECR
Frontend Image
Backend Image
(Latest: sha-fa37c6a)

5. Push Images

Route 53
DNS
(Optional)

VPC (10.0.0.0/16) - Multi-AZ Deployment

Public Subnets (3 Availability Zones)

us-east-1a (10.0.1.0/24)

Public
Subnet 1a

us-east-1b (10.0.2.0/24)

Public
Subnet 1b

us-east-1c (10.0.3.0/24)

Public
Subnet 1c

NAT GW
us-east-1a

NAT GW
us-east-1b

Application
Load Balancer
(ALB)

NAT GW
us-east-1c

Path Routing:
/* → Frontend
/api/* → Backend

Internet Traffic

/* → Frontend

/* → Frontend

Outbound

Outbound

/* → Frontend

/api/* → Backend

/api/* → Backend

/api/* → Backend

Outbound

Private Subnets (3 Availability Zones)

Amazon EKS Cluster (Kubernetes v1.30)

EKS Worker Nodes (Auto Scaling: 3-10 nodes)

Backend Tier (Application Layer)

Backend Pods (3 replicas, HPA: 3-10)

Frontend Tier (Web Layer)

Frontend Pods (3 replicas, HPA: 3-10)

us-east-1a (10.0.11.0/24)

us-east-1b (10.0.12.0/24)

us-east-1c (10.0.13.0/24)

Internet
Gateway

Private
Subnet 1a

Private
Subnet 1b

Private
Subnet 1c

EKS Control
Plane
(Managed)

Frontend Pod 1
React 18.2.0
Nginx
Port: 80
CPU: 500m
RAM: 512Mi

Frontend Pod 2
React 18.2.0
Nginx
Port: 80
CPU: 500m
RAM: 512Mi

Frontend Pod 3
React 18.2.0
Nginx
Port: 80
CPU: 500m
RAM: 512Mi

Worker Node 1
t3.medium
2 vCPU, 4GB
us-east-1a

Worker Node 2
t3.medium
2 vCPU, 4GB
us-east-1b

Backend Pod 1
Flask 2.3.2
Gunicorn
Port: 5000
CPU: 1000m
RAM: 1Gi

Backend Pod 2
Flask 2.3.2
Gunicorn
Port: 5000
CPU: 1000m
RAM: 1Gi

Backend Pod 3
Flask 2.3.2
Gunicorn
Port: 5000
CPU: 1000m
RAM: 1Gi

Worker Node 3
t3.medium
2 vCPU, 4GB
us-east-1c

AI Agents:
• Intent Detector
• Recommendation
• Order/Shipping
• Fraud Detection

Upload Images

Upload Images

IAM Auth

Fetch Secrets

Fetch Secrets

Upload Images

Logs & Metrics

Logs

Logs

Fetch Secrets

MongoDB Protocol
Port: 27017

MongoDB Protocol
Port: 27017

MongoDB Protocol
Port: 27017

Query Cards

Store Reports

CRUD Orders

Database Tier (Data Layer)

MongoDB StatefulSet (1 replica)

MongoDB Pod
MongoDB 6.0
Port: 27017
CPU: 500m
RAM: 1Gi

Collections:
• users
• cards
• orders
• defects
• fraud

Persistent Storage

Amazon EBS
Volume
gp2: 20GB
Encrypted
/data/db

AWS Supporting Services

Storage Services

Amazon S3

Buckets:
• Defect Images
• Fraud Receipts
• Backups
• Logs

Security & Access Management

AWS IAM

Roles:
• EKS Cluster Role
• Node Instance Role
• Pod Service Accounts
• GitHub Actions User

AWS Secrets
Manager

• OpenAI API Key
• MongoDB Creds
• ECR Tokens

Monitoring & Logging

Amazon
CloudWatch

Log Groups:
• EKS Cluster
• Application
• Performance

Trigger Alerts

CloudWatch
Alarms

• CPU > 80%
• Memory > 85%
• Pod Failures
• 5xx Errors

Notify

Notifications

Amazon SNS

Topics:
• Deployments
• Alerts
• Errors

Multi-Agent AI System (LangChain)

Main Agent
Intent Detector
(GPT-4)

AI Requests

AI Requests

AI Requests

Route Intent

Route Intent

Route Intent

Route Intent

Specialized Agents

Recommendation
Agent

Task:
• Query Generation
• Card Matching
• Personalization

Defect Detection
Agent

Task:
• Image Analysis
• Severity Rating
• Replacement

Order/Shipping
Agent

Task:
• Order Creation
• Tracking
• Cancellation

Fraud Detection
Agent

Task:
• OCR Processing
• Risk Analysis
• Card Blocking

API Calls

Orchestration

Generate Recs

Analyze Defects

Store Reports

Analyze Fraud

Extract Text

External AI/ML Services

OpenAI GPT-4
API

Models:
• GPT-4 Turbo
• GPT-4 Vision
• Embeddings

Use Cases:
• Intent Detection
• Recommendations
• Defect Analysis
• Fraud Detection

Tesseract OCR

Version: 0.3.10

Use Case:
• Receipt Text
Extraction
• Fraud Detection

LangChain
Framework

Version: 0.0.267

Features:
• Multi-Agent
• Orchestration
• Memory
• Tools

Credit Card Recommendation System - Complete AWS Architecture
(All Services, Infrastructure, CI/CD, AI Agents, Security)